

▪ Residência unifamiliar

▪ Cargas

⇒ Permanentes:

↳ Peso próprio: $2,07 \text{ kN/m}^2$

↳ Revestimento: $0,50 \text{ kN/m}^2$

⇒ Variável: $1,5 \text{ kN/m}^2$

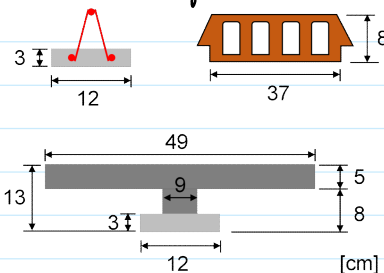
⇒ LT 13 (8+5)

⇒ Tabela H8 – $8 \times 37 \times 20$

⇒ Concreto C25

↳ Agregado basalto

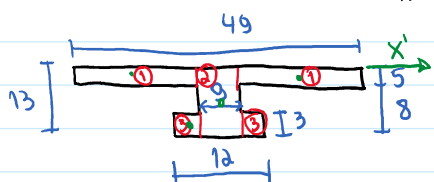
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Clas de cálculo: $3,85 \text{ m}$, simplesmente apoiada

ELS de Deslocamento Excessivo

$$E_{cs} = 28980 \text{ MPa} = 2,898 \cdot 10^7 \text{ kPa}$$



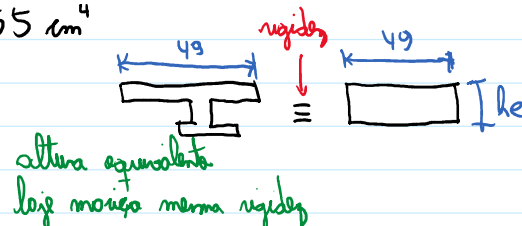
$$y_{cg} = \frac{S_x}{A} = \frac{\sum y_i A_i}{A} = \frac{2,5 \cdot (49-9) \cdot 5 + 6,5 \cdot 9 \cdot 13 + 11,5 \cdot 3 \cdot 3}{(49-9) \cdot 5 + 9 \cdot 13 + 3 \cdot 3} = \frac{1364}{326} = 4,18 \text{ cm}$$

$$I_x = \frac{(49-9) \cdot 5^3}{3} + \frac{9 \cdot 13^3}{3} + \frac{3 \cdot 3^3}{12} + 3 \cdot 3 \cdot 11,5^2 = 9455 \text{ cm}^4$$

$$I_{x, cg} = 9455 - 326 \cdot 4,18^2 = 3759 \text{ cm}^4$$

$$3759 = \frac{49 \cdot h_e^3}{12}$$

$$h_e = \sqrt[3]{\frac{3759 \cdot 12}{49}} = 9,73 \text{ cm}$$



$$D_e = \frac{E_{cs} h^3}{12(1-\nu^2)} = \frac{2,898 \cdot 10^7 \cdot 0,0973^3}{12(1-0,2^2)} = 2317 \text{ kNm}$$

$$\text{Bem 7: } l_x = 3,85 \text{ m}$$

$$f_{d, m} = g_k + \psi_2 q_k = 2,57 + 0,3 \cdot 1,50 = 3,02 \text{ kN/m}^2$$

$$w_0 = \frac{5}{384} \frac{f_{d, m} l_x^4}{D} = \frac{5}{384} \cdot \frac{3,02 \cdot 3,85^4}{2317} = 3,73 \cdot 10^{-3} \text{ m} = 3,73 \text{ mm}$$

$$w_{ad} = (1+\phi) w_0 = (1+2,5) \cdot 3,73 = 13,05 \text{ mm}$$

$$w_{adm} = \frac{l}{250} = \frac{3850}{250} = 15,4 \text{ mm}$$

Como $w_{ad} = 13,05 \text{ mm} < w_{adm} = 15,4 \text{ mm}$,
a laje possui rigidez adequada

Estado Limite Último

Dimensionamento à flexão

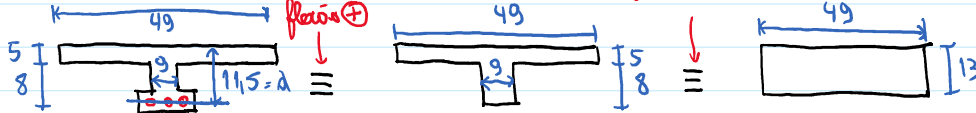
Armadura mínima

$$A_{s,min} = \rho_{min} A_c = \frac{0,15}{100} \cdot 326 = 0,489 \text{ cm}^2/\text{vig}$$

Momentos fletores:

$$f_d = 1,4 \cdot 2,57 + 1,4 \cdot 1,50 = 5,70 \text{ kN/m}^2 \quad \times 0,49 \text{ m/vig}$$

$$M_d = \frac{f_d l_x^2}{8} = \frac{5,70 \cdot 3,85^2}{8} = 10,56 \text{ kNm/m} \quad 5,17 \text{ kNm/vig} = 5,17 \text{ kNm/vig}$$



Supondo a seção retangular com $b = 49 \text{ cm}$

$$\lambda x = \frac{d}{\lambda} \left(1 - \sqrt{1 - \frac{2 M_d}{\alpha_c \eta_c f_{cd} b d^2}} \right) = \frac{11,5}{0,8} \left(1 - \sqrt{1 - \frac{2 \cdot 5,17}{0,85 \cdot 1,0 (2,5/1,4) \cdot 49 \cdot 11,5^2}} \right) = 0,78 \text{ cm}$$

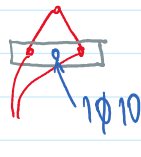
$$\lambda x = 0,8 \cdot 0,78 = 0,62 \text{ cm} < h_f = 5 \text{ cm} \rightarrow \text{hipótese correta}$$

$$A_s = \frac{\alpha_c \eta_c f_{cd} b \lambda x}{f_{yd}} = \frac{0,85 \cdot 1,0 \cdot (2,5/1,4) \cdot 49 \cdot 0,8 \cdot 0,78}{43,48} = 1,07 \text{ cm}^2/\text{vig}$$

TR 8644



TR 8645



Ø4,2

2Ø8

Ø5

1Ø10

2Ø8 + 1Ø5

Esforço constante

$$\text{Soluções: } R_d = \frac{f_d l_x}{2} = \frac{5,70 \cdot 3,85}{2} = 10,97 \text{ kNm/m} \quad 5,38 \text{ kN/vig}$$

$$R_{rd1} = G_{rd} K (1,2 + 40 \rho_1) b_w d = 0,032 \cdot 1,485 (1,2 + 40 \cdot 0,0116) \cdot 9 \cdot 11,5 = 8,18 \text{ kN/vig}$$

$$G_{rd} = 0,25 f_{ctd} = 0,25 \cdot 0,15 f_{ctk}^{2/3} = 0,25 \cdot 0,15 \cdot 25^{2/3} = 0,32 \text{ MPa} = 0,032 \text{ kN/cm}^2$$

$$K = 1,6 - d = 1,6 - 0,115 = 1,485$$

$$2Ø8 + 1Ø5 \quad A_{s1} = 1,20 \text{ cm}^2$$

$$\rho_1 = \frac{A_{s1}}{b_w d} = \frac{1,20}{9 \cdot 11,5} = 0,0116$$

$$V_{sd} = 5,38 \text{ kN/vig} < V_{rd1} = 8,18 \text{ kN/vig}, \text{ a larg. possui segurança adequada para o esforço constante}$$

Armadura de distribuição

$$\text{NBR 14860-1} \quad A_{s,dist} = 0,60 \text{ cm}^2/\text{m}$$

EF-96

$$A_{s,dist} = 5 \cdot \frac{h_f}{f_{yd}} = \frac{5 \cdot 5}{43,48} = 0,57 \text{ cm}^2/\text{m}$$

$$A_{s,dist} = 0,60 \text{ cm}^2/\text{m}$$

Ø4,2 c/23 cm

Ø5 c/32 cm